

What does the yield curve say? An exercise into Yield Curve estimation

Kuldeep Kachhwaha Nidhi Srivastava Soham Shevade

May 25, 2021

Abstract

Charles Nelson and Andrew Siegel proposed a model in 1987 which is considered to be gold standard for estimating the term structure of interest rates over time. They showcased that their model was flexible enough to capture the various possible shapes of the yield curves. The main focus of our study is to conduct an exercise into estimation of Yield Curves by fitting Nelson Siegel Model using OLS to the calculated yield to maturities. We have used bond price data published by NSE for this estimation exercise. Further in our study we have also tried to identify any structural changes in the economic activity, especially focusing on time period between 2006-2008 and 2019-2021 in the context of Indian Economy. A broad analysis of the yield curves within the time frames under consideration tells us that they are more or else moderately steep and pretty stable. In general, we were able to see that the general shape of the Yield Curves that we calculated was more or less unaffected by the volatile economic scenarios of 2006-2008 and 2019-2021. Thus, we loosely reckon that Yield Curves are not that powerful of an indicator (to predict economic recessions and crisis) in Indian context as it is for the US or European economy.

1 Introduction

The financial market is the key channel of transmission of monetary policy into the real economy. The central banks all around the world have increasingly started using Yield Curves as a signal to supervise the market's functioning. In order to improve the precision of policy action, it has become all the more important to get a spot on estimation of Yield curves. Yield Curves are the fundamental building block of finance space. They have wide applications in cash flow discounting and market sentiment analysis. At an initial glance, it is quite easy to grasp the theoretical aspects of yield curve. However, what we wanted to attain through this project is a first hand experience of estimating the curves using real time data. At a primitive stage, one would just see it as a graphical representation of time to maturity and yield rates. However, once we started our journey of actually collecting the data, we were exposed to the intricacies of the data set wherein there were tonnes of other variable that demanded equal attention from our end. For instance, one would naively never give a thought to price volatility as a factor affecting the quality of the data set. The data exploration process was extensive which posed a lot of technical challenges that we actually never anticipated. Additionally, the project exposed us to the subjectivity of the financial data analysis process involving a lot of decision making. This gave us a perspective into why it might be that, from the same information and data sets available, different analysts come to polar opposite conclusions regarding buying or selling of securities. In general, we got a whole new appreciation for dealing with financial data. We have also tried to simplify the estimation process for a better understanding for people of the same academic calibre.

The aim of the study is to conduct a step by step exploration of yield curves and its estimation using Nelson Siegel technique. We have used NSE bond price data base which has all the necessary variables such as the coupon rates, maturity dates, frequency of trades, value weighted trade price, etc. We move further with an intent to investigate the shapes and forms of yield curves during the time frame under consideration. We are also interested in seeing if yield curves can indicate impending economic slowdowns and crisis in Indian context as there is a lot of discussion about yield curves being very good predictors of impending economic crisis in the US economy. We also to see if we can identify any structural changes in the economic activity, just by looking at the yield curve. Finally, we also look at if there are any identifiable changes or shifts in the yield curve because of Covid-19.

2 What are yields?

A bond's yield is the internal rate of the payment structure. In other words, the interest rate at which the present value of the stream of payments expected from the bond is exactly equal to the current price of the bond. This is also called as Yield to Maturity(YTM)[4]. Let us suppose that F is the face value of the bond and C is the coupon payment associated with it. There are n periods remaining. Suppose that the bond is trading at price P in the secondary market, then the yield to maturity λ is given by

$$P = F/(1 + \lambda)^n + \sum_{i=1}^n C/(1 + \lambda)^i \quad (1)$$

The value of λ is the yield to maturity, which is the interest rate implied by the bond. The first term is the present value of the face value of the bond, whereas the second term addresses the present value of all of the coupon payments for the remaining k years. The sum of both the terms is set equal to the price of the bond and we solve for λ to get the YTM for a given bond.

The bond price formula can also be used to obtain a qualitative understanding of governing factors like price, yield, coupon and time to maturity. An understanding of such relationship is crucial to get a deeper understanding into bond portfolio construction. In general, the yield to maturity of the bonds are constantly moving. This is the result of constant fluctuations in the bond prices in the secondary market. Looking at the formula, it is easy to see that, if the price moves, the yields change and vice-versa.

In general, there exists an inverse relationship between price and yield to maturity of the bond. If the price of a bond goes down, the yield to maturity goes up and vice versa. The relationship between yield to maturity and price in fact goes through the channel of demand. Yield goes down when the demand for that bond rises and Yield goes up if the demand for a bond falls. As we can see, this is a stable i.e. self correcting equilibrium. It is common for the investors to converse in yield to maturity terms, as yield gives a holistic approach of not only comparing bonds with different maturities but also with saving accounts, inflation and income from other classes of assets. The bond continues paying income even if the price fluctuates, so if the price rise, this might mean that the investor would earn capital gains in addition to the income by trading it. If the price falls, he faces a capital loss.

The time to maturity and the coupon rate also has a considerable influence on the price-yield relationship of the bond. If a higher coupon is offered on the fixed stream of payments, then the price for the bond naturally goes up. At the same time, longer maturities are subjected to higher price-yield sensitivities. Investors holding longer term maturities bonds are more prone to the direct interest rate risk. Clearly, any change in the yield brings about an impact on the price of the bond. This explicitly affects the near-term value of the bond. Thus, if the investor wants to trade his bond before maturity, the price is calculated using the price-yield curve.

3 Yield Curves

To get a comprehensive picture of the financial market, we take into account the Yield Curves. Yield Curves are a locus of all yield of bonds of same quality class as a function of time to maturity. It is a snapshot taken at a particular point of time. In general, the variation in bond yields differ for different quality class. A high quality bond would trade with a higher price, whereas low quality bonds are marked down. Moreover, the yields are also subjected to variation in time to maturities. Bonds with more time to maturity have higher yields and vice-versa. The yields gradually rises with rising time to maturity, giving the curve a smooth upward sloping shape. However, the yield-curves are not always upward sloping, they can take various shapes. If, for instance, the longer maturities bonds take up lower yields than shorter maturity bonds, it results in an inverted yield curve. It indicates that the shorter maturity bond yields are increasing rapidly, and the long maturity bond yields are not changing much because the investor believe that the rise is temporary.

The generic shape of the yield curve is explained by various hypotheses[1]. These are as follows:

3.1 The Expectation Hypothesis

The Expectation hypothesis propounds that the long term interest rates are an average of short term interest rates that is expected to prevail during the life of the bond. This theory assumes that the various series of short term securities are perfect substitutes of one another and the investors are risk-neutral. Thus, according to this theory, the various shapes of the bond at a particular point of time is due to varying expectations of future one-year rates by the investors. If the rates are expected to rise, it would lead to an upward sloping curve, flat when not expected to rise, downward sloping when expected to fall. Historical analysis suggests a rising yield curves, implying that they always expect the rates to rise. Thus, this hypothesis closely fails to explain the shape of the yield curve.

3.2 The Segmented Market Hypothesis:

This hypothesis clubs the bonds with market of similar maturities into one market. In each segregated market, the yields are determined by the intersection of supply and demand. The primary proposal is that bonds with different maturities cannot be a perfect substitute for one another, thus abandoning the basic assumption of our previous hypothesis. The rationale instead is that the investor according to his financial needs would target the market with specific maturities. If a person has idle cash which has no immediate requirement, he/she would prefer to lock in the funds in longer maturity markets. The person would be more comfortable in buying a long term securities than buying short term at multiple point of time, as it would minimise transaction costs and reduce effort and interest rate risks. Given a demand for loanable funds, when the supply of short term loanable funds is greater than the supply of long term loanable funds, short term rates will be lower than long term rates, giving an upward sloping curve. In contrast, for a given demand for loanable funds, if the supply of short term is less than long term, the short term rates would be higher than long term rates. This would give a downward sloping yield curve. Even though the theory explains the shape of the curve, it does not give any information about the changes in the pattern of the yield. Secondly, in reality the long term and short term rates show empirical relationship, abandoning the assumption of both being independent of one another.

3.3 The Preferred Habitat Hypothesis:

This hypothesis is a blend of key assumptions of two hypothesis stated above. It says that the investors do have preferences for the segmented market, but they also willing to substitute

away from their preferred terms if they are compensated for doing so. This compensation is known as the Term premium. This theory also gives a fair share of recognition to the short term and long term rates moving together. This can be better explained mathematically as follows: Let's consider a model with two time periods. The term premium is given by $\alpha = 0.01$. The interest rate for one year bond is $i_1 = 0.06$. The interest rate for two year bond is given by

$$i_2 = 0.01 + ((0.06 + E(i_1))/2) \quad (2)$$

Thus, if the investor expects the short term rate to rise substantially, we get an upward sloping yield curve. If the short term rate is expected to fall moderately, we get a flat yield curve. And if it is expected to fall dramatically, we get a downward sloping curve. This implies that unless there is dramatic change in expectation, the yield curves would generally stay stable and rising. Taken everything together, the hypothesis fits very well with the empirical data.

4 Data and Methodology

To facilitate the estimation of term structure, we have considered a parametric approach used in the Nelson- Seigel model. A parametric model is defined to be a model in which all information is represented within its parameters. In 1987, Nelson and Seigel introduced the idea that by constraining the zero coupon rate to be a special function of time to maturity with enough free-to-choose parameters, that could actually produce a good fit to all the market curves by choosing those parameters. The function that we fit our sample data using OLS is given by:

$$f(t) = \beta_0 + \beta_1 \exp(-t/\tau_1) + \beta_2 (t/\tau_1) \exp(-t/\tau_1) \quad (3)$$

To understand how the behaviour of this curve changes with respect to changes in parameters, we start by looking at 2 basic functions which are:

- $f_1(t) = \exp(-t/\tau)$
- $f_2(t) = (t/\tau) \cdot \exp(-t/\tau)$

The first thing that we can notice is that τ is a scale parameter. Higher values of τ will stretch both the graphs horizontally. Negative values of τ will flip these graphs around the y axis. So negative τ isn't really something which we hope to observe.

The first function is a familiar one. It is an exponential decay function and it is very easy to see how it behaves. The second function is indeed a more interesting one. This has a very counter-intuitive shape. Looking at first and second order derivatives makes things a lot easier.

$$\begin{aligned} \frac{\partial f_2(t)}{\partial t} &= \frac{e^{-\frac{t}{\tau}} (\tau - t)}{\tau^2} \\ \frac{\partial^2 f_2(t)}{\partial t^2} &= \frac{e^{-\frac{t}{\tau}} (t - 2\tau)}{\tau^3} \end{aligned}$$

Looking at the first derivative, it is clear that we are going to have a critical point at $t = \tau$. Looking at the second derivative it is apparent that the second derivative changes sign at the point $t = 2\tau$. When t is less than 2τ and τ is positive, the function will be concave as the second derivative will be negative. Point 2τ is a point of inflection as that is where the function changes from concave to convex.

Now that we understand these 2 functions, it is much easier to look at the NS model which is going to be used to fit a yield curve to our calculated yield to maturities. The NS

model is just a linear combination of the two functions that we discussed earlier. It has an intercept term β_0 which will shift the curve vertically. We can think of β_1 and β_2 as weights assigned to f_1 and f_2 .

In order to fit the model well, it is important that we use a reliable data set. We have followed methodology described in [5]¹ For our study, we have taken data from NSE Wholesale market data base. The primary bond instrument considered is the Government Loans(GS), as they are characterized as AAA Bonds(as per CRISIL Credit Ratings), meaning they have close to zero risk of default.This is also to meet the requirement of same credit classes while structuring yield curves. The time frame under consideration is 2001-2002, 2006-2008 and 2019-2020.In order to cover a large range of years, monthly data for bonds were considered. To prepare our bond sample, the following characteristics were studied and analyzed:

- A good measure for liquidity of bonds is turnover values or bid-ask spread. However, due to unavailability of the data we have mainly considered the count of number of times a particular bond is traded. The assumption made here is that more liquid the market is, higher the number of trades are conducted.
- Volatility - To bring uniformity to the bond sample, price volatility of all securities was considered. This was measured using standard deviation as well as visualized using line graphs. Securities with extreme volatility were removed from the sample.
- Selection of maturity spectrum- To bring standardisation to the bond sample,residual maturity spectrum ($1year \leq t \leq 20years$) was considered.
- Missing Values/Outliers - There were many securities that are traded infrequently, hence do not add any significant information to the bond data set. Thus, they were removed.

The elementary data analysis was conducted using MS Excel. The process followed was threefold. First, we downloaded the data from NSE-WDB using the code "GC". Then, data was filtered month-wise. Secondly, to understand the characteristics of price and yields-security name wise, we created line graphs(maturity wise filter) and standard deviation measures of price were used. Accordingly, particular security names were selected, keeping in mind the frequency of trades and missing information. Thirdly, we created another column called as "Model Filter" wherein the particular securities selected were marked as 1 and others as 0. This was followed by keeping the file path in our Python Code as variable. When the data is entered,the code starts with pre-processing the given data.We wrote a function in the code to specifically calculate the yield to maturities,which further was used to estimate the NS function using OLS. Finally the code iterates over all the months and gives us one yield curve per month. The code is reproducible and commented.

5 Results and Discussion

Using the high quality data set, the Nelson-Siegel estimation technique was applied. The results and findings are presented below. Fig. 1 to 5 are the yield curves that we have reached at. In general, the slopes of the yield curve give us the value of instantaneous forward rates, hence add a lot of information about the changes in the interest rate or expectations. Thus, the spread and shape tell us a lot about the twist and turns in the economic activity. The normal observable shapes were mainly - Upward sloping, flat and rarely downward sloping. In absolute sense, an upward sloping implies the usual that the long term rates are higher than the short term rates. Besides that it also tells us that due to higher yields for long term

¹"Yield curve modelling and a conceptual framework for estimating yield curves: evidence from the European Central Bank's yield curves," Statistics Paper Series 27, European Central Bank.

bonds, investors would expect the bonds value to be lower. As a result of which, they tend to park in their excess funds in short term bonds and they are reluctant in investing their money in the long term. This is a sign of good liquidity in the financial market. Sometimes, if the curves are too steep, they also imply a higher expectation of inflation in the future. The rationale is that higher inflation would imply lower value of the bonds and thus higher yields. A higher expectation of inflation also supports growth in the economy. It is pretty apparent to expect an upward sloping curve, as investors normally expect a higher premium for locking their funds for long time.

On the other hand, a downward sloping/inverted curve are rare, yet striking. An inverted curve would imply that long term yield are lower than short term yields. This would mean that the investors would expect a higher value for their bonds in the future, thus would withdraw their funds from short term and transfer them to long term bonds. This has a direct impact on liquidity of the market, hence not much welcomed by the policy makers. A flat yield curve is another prominent shape that came in notice. This shape usually occurs when the spread between the short term and long term bonds are not much, indicating that the economy is in transition.

A broad analysis of the yield curve within the time frames under consideration tells us that they are more or else moderately steep and pretty stable. As explained using the Preferred Habitat Hypothesis, unless there are any dramatic changes expected, the yield curves are generally stable. So, the output gives us a hard time to draw out any out of the ordinary conclusions. In general, we can say that since the data set pertaining to India is unaffected by volatile and dynamic aspects of the bonds, we have derived pretty stable results. Thus we loosely reckon that Yield curves are not that powerful as an indicative instrument in the Indian context.

1 to 3 give us a wide coverage of the yield curve shapes during the Financial Crisis of 2007-08. Preceding the crisis, the yield curves of the worst affected countries did signal an upcoming recession. That essentially contributed in making the yield curve much of a phenomena, especially in the United States. However, our analysis of the crisis period gives us steady results for India. Few of the curvatures noticeable in 1i, 2a,3g are owing to very erratic behaviour of extremely short maturities(mostly less than 1 year). Thus, the curvatures do not convince us of any deflecting observation in our conclusion. In other words, we do not have a solid reason to conclude that Yield curve signal something unusual about the crisis hitting India.

4 to 5 give us a monthly snapshot of the Indian bond market during the Covid-19 pandemic crisis. Again, the curves for 2019 are mildly steep, meaning they are not too steep nor are they absolutely flat. The curves for 2020 have conspicuous upward sloping/steep curve. It is noticeable that the spread between rates of short term and long term bonds have considerably increased from 4 to 5. Using this observation, we deduce that the yields for long term bonds have risen from 2019 to 2020. Theoretically, higher yields imply that the investors would expect the value of bonds to be lower due to fall in prices. Thus, they are placing higher demands on short term bonds, increasing its price and lowering its yields. In expectation that the situation would subside at some point, they would park more funds in short term bonds and use them in the later future. Thus we get a satisfactory upward sloping curve in 2020.

6 Recent Developments

Yield Curve Controlling, as the name suggests, is a tool used by the Central banks to so called "Flatten the Curve". This has been in discussion all over the Monetary Policy space as a happening thing. Even in the United States, the investors and stakeholders have started placing bets on stock market in anticipation of the YCC policy[6]. The Fed is continually

purchasing assets especially Treasury Bonds, creating an impact on rates and expectation. Another policy instrument in talks is Forward Guidance. Under this, the Fed confidently communicates their future policy action and forecasts to influence the market.

In India, RBI also has been heavily buying and selling government securities under Operation Twist[2], with an aim to lower long term yields and lower the spread of the curves. This is important to fight any impending recession. The outputs for 2020, clearly has no indication of yield curves flattening, rather they are steeper than 2019 yield curves. One reason could be even further falling of long term yields, due to increasing demands of short term bonds. As the lockdown eases, firms and people are in dire need of immediate cash flows and are placing higher bets on short term bonds. One positive angle to this would be that the RBI is able to maintain liquidity in the market pretty well. This is further accompanied by lower borrowing costs for firms and cooperates. As of 7 April 2021, the RBI has announced that it will Rs 1 Trillion worth of Government Bonds. This is what we perceive as Forward Guidance action taken by the RBI and trying its best to "Flatten the Curve".[3]

References

- [1] Michael R Baye and Dennis W Jansen. *Money, banking, and financial markets: An economics approach*. Houghton Mifflin College Division, 1995.
- [2] Yogesh Dayal. Rbi press releases governor's statement, february 5, 2021. Feb 2021.
- [3] Aparna Iyyer. bend-it-like-rbi-das-yield-curve-management-takes-centre-stage. *The livemint*, Apr 2021.
- [4] David G Luenberger et al. Investment science. *OUP Catalogue*, 1997.
- [5] Per Nymand-Andersen. Yield curve modelling and a conceptual framework for estimating yield curves: evidence from the european central bank's yield curves. Technical report, ECB Statistics Paper, 2018.
- [6] Bhargavi Sakhtivel and Nikita Singh. What the hell is a yield curve, why would anyone want to control it and other annoying questions answered. *Moneycontrol*, Aug 2020.

7 Python Code

```
import pandas as pd
import numpy as np
from nelson_siegel_svensson.calibrate import calibrate_ns_ols
import matplotlib.pyplot as plt
import calendar

def tpv(fv, coupon, years, DiscountingRate):

    # fv = Face value of a bond
    # coupon = Coupon rate in percentage (without percent sign)
    # years = years to maturity
    # DiscountingRate = Rate of discounting in decimal format
    # and not in percent format

    if coupon > 0:

        years = int(years)

        tpv = 0

        for year in range(1, years+1):

            tpv += coupon / ((1 + DiscountingRate)**year)

        tpv += fv / ((1 + DiscountingRate)**years)

        return tpv

    else:

        tpv = fv / ((1 + DiscountingRate)**years)

        return tpv

def ytm(p, cr, y, fv=100):

    # p = current price
    # fv = face value of the bond
    # cr = coupon rate in percent
    # y = years to maturity

    cr = cr / 100.0

    coupon = fv * cr

    ytm = cr

    condition = True
```



```

while condition:
    if (p < fv):
        ytm += 0.00001
    else:
        ytm -= 0.00001

    pv = tpv(fv, coupon, y, ytm)

    if (p < fv):
        condition = pv > p
    else:
        condition = pv < p

return ytm

# Bond Data preprocessing

BondDataFilePath = "01-01-2020-TO-30-12-2020GS.csv"

dfBondData = pd.read_csv(BondDataFilePath)

dfBondData = dfBondData[dfBondData["Filter"]==1]

dfBondData['Date'] = pd.to_datetime(dfBondData['Date'])

dfBondData["Month"] = pd.DatetimeIndex(dfBondData['Date']).month

dfBondData["Year"] = pd.DatetimeIndex(dfBondData['Date']).year

dfBondData["Maturity_Year"] = dfBondData["Security_Name"].str[2:6].astype(int)

dfBondData["Res_Maturity"] = dfBondData["Maturity_Year"] - dfBondData["Year"]

dfBondData["Res_Maturity"] = dfBondData["Res_Maturity"].replace(0,1)

dfBondData["Coupon_Rate"] = dfBondData["Issue_Name"].str.replace("%","")

dfBondData["Coupon_Rate"] = dfBondData["Coupon_Rate"].astype(float)

dfBondData = dfBondData[["Date","Month","LTP","Res_Maturity","Coupon_Rate"]]

dfBondData["Res_Maturity"] = dfBondData["Res_Maturity"].astype(float)

lstmonths = dfBondData["Month"].unique()

# End of preprocessing and starting with Calculation of "yield to maturities"
# and yield Curve estimation for each month in the dataset

for month in lstmonths:

    dfMonthData = dfBondData[dfBondData["Month"]==month]

```

```

dfMonthData = dfMonthData.drop_duplicates()

dfMonthData["Calc_YTM"] = dfMonthData.apply(lambda row : ytm(row["LTP"],
row["Coupon_Rate"], row["Res_Maturity"])), axis = 1)

time = dfMonthData["Res_Maturity"].to_numpy()

calcytm = dfMonthData["Calc_YTM"].to_numpy()

curve, status = calibrate_ns_ols(time, calcytm, 1.0)

print(curve)

monthname = calendar.month_name[month]

stryear = dfBondData.iloc[0,0].year

t = np.linspace(1,30,60)

fig, ax = plt.subplots( nrows=1, ncols=1 )

ax.plot(t, curve(t))

plt.xlim(0, 30)

plt.ylim(0.03, 0.12)

plt.title("Yield_curve_for_"+monthname+"_"+str(stryear))

plt.xlabel('Year')

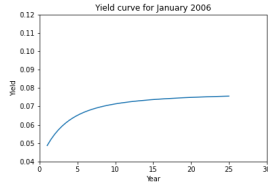
plt.ylabel('Yield')

fig.savefig(str(month)+".png")

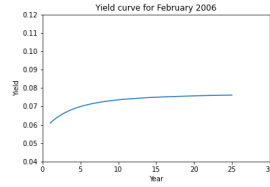
plt.close(fig)

```

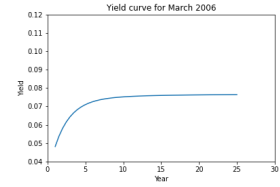
8 Figures



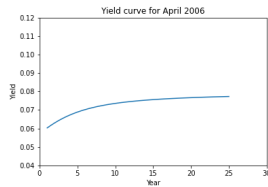
(a) January 2006



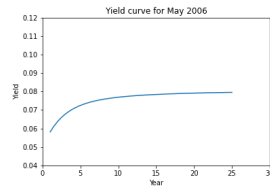
(b) February 2006



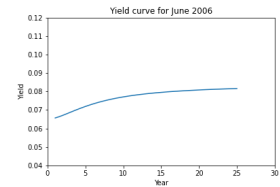
(c) March 2006



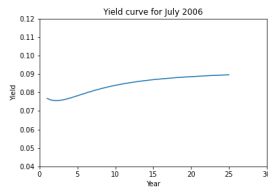
(d) April 2006



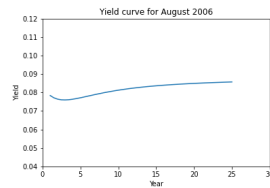
(e) May 2006



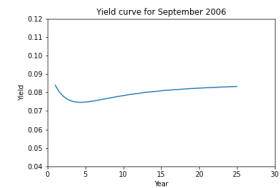
(f) June 2006



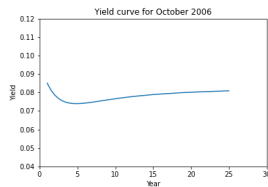
(g) July 2006



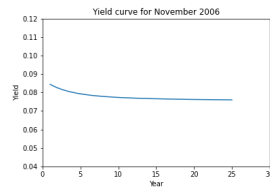
(h) August 2006



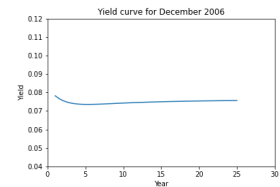
(i) September 2006



(j) October 2006

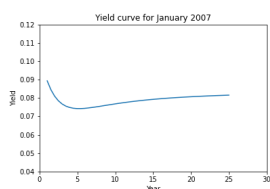


(k) November 2006

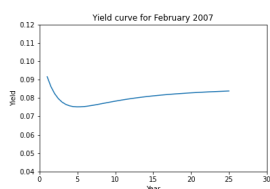


(l) December 2006

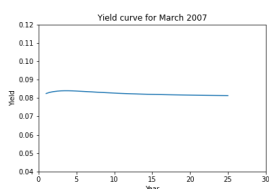
Figure 1: 2006



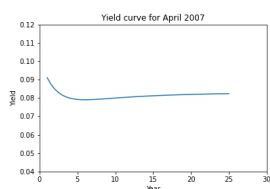
(a) January 2007



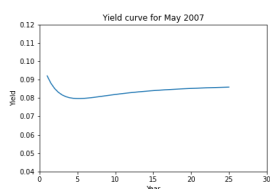
(b) February 2007



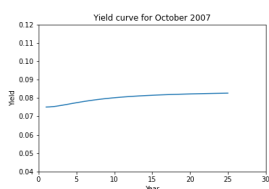
(c) March 2007



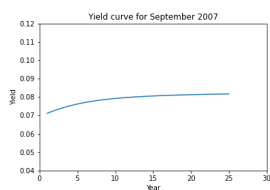
(d) April 2007



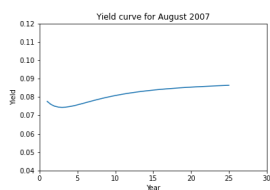
(e) May 2007



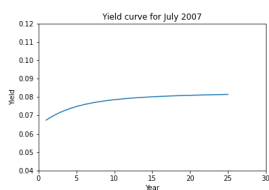
(f) June 2007



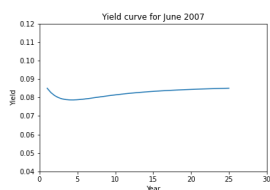
(g) July 2007



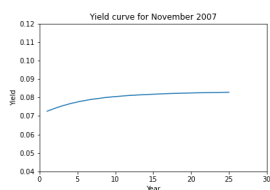
(h) August 2007



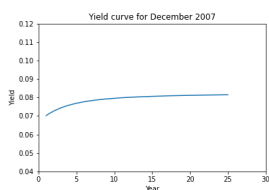
(i) September 2007



(j) October 2007

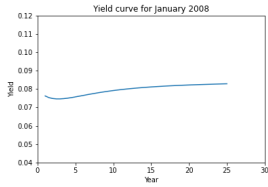


(k) November 2007

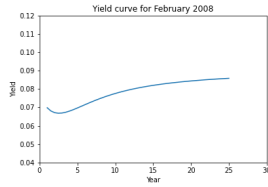


(l) December 2007

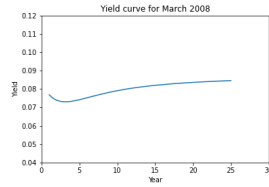
Figure 2: 2007



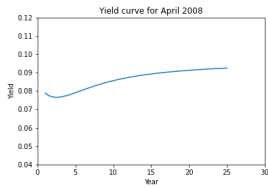
(a) January 2008



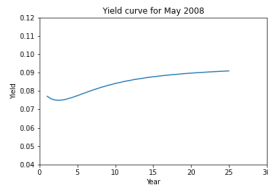
(b) February 2008



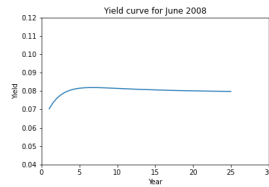
(c) March 2008



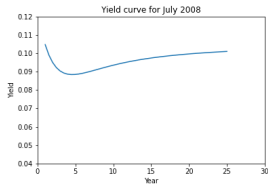
(d) April 2008



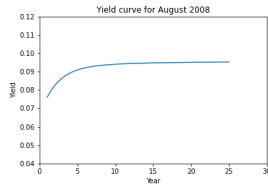
(e) May 2008



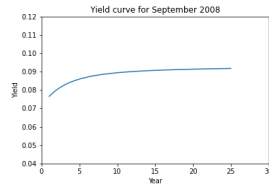
(f) June 2008



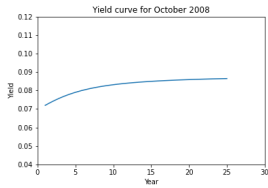
(g) July 2008



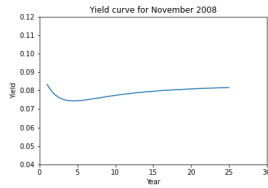
(h) August 2008



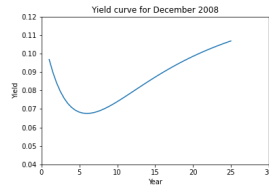
(i) September 2008



(j) October 2008

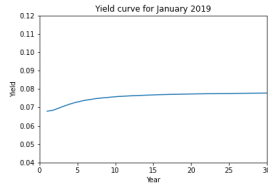


(k) November 2008

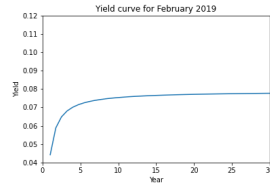


(l) December 2008

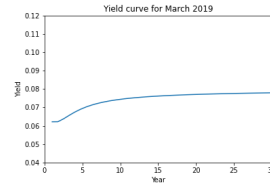
Figure 3: 2008



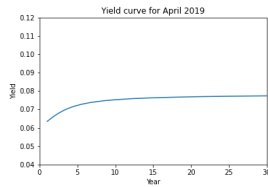
(a) Januar 2019



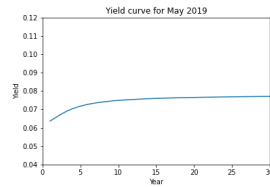
(b) February 2019



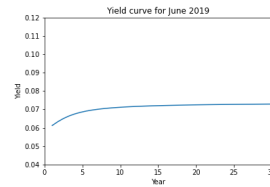
(c) March 2019



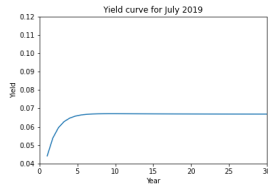
(d) April 2019



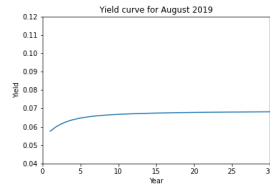
(e) May 2019



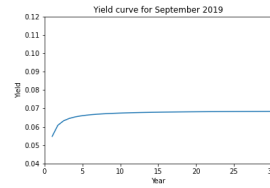
(f) June 2019



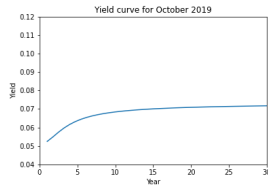
(g) July 2017



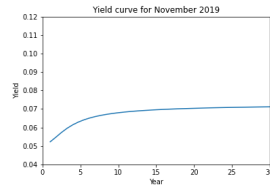
(h) August 2019



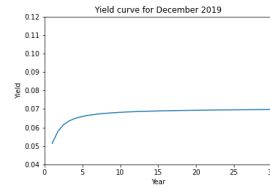
(i) September 2008



(j) October 2019

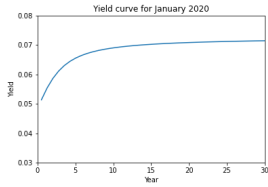


(k) November 2019

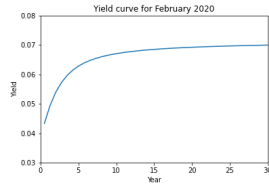


(l) December 2019

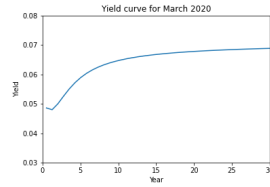
Figure 4: 2019



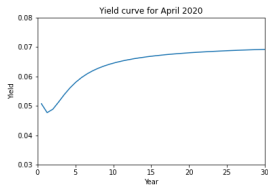
(a) January 2020



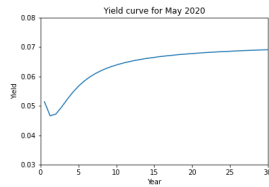
(b) February 2020



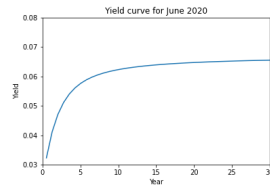
(c) March 2020



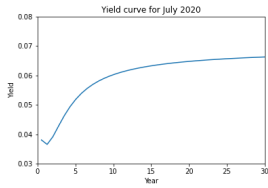
(d) April 2020



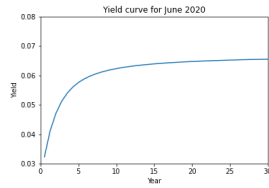
(e) May 2020



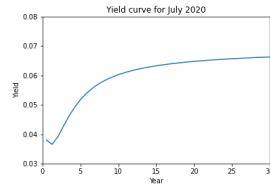
(f) June 2020



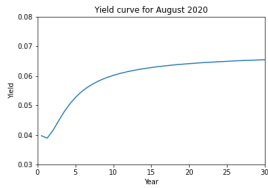
(g) July 2020



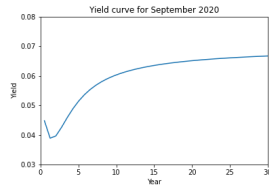
(h) August 2020



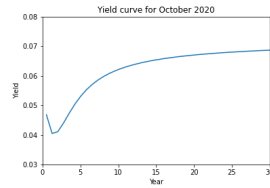
(i) September 2020



(j) October 2020



(k) November 2020



(l) December 2020

Figure 5: 2020